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COUSE CODE: SBIT322

LEVEL: 300

ASSIGNMENT 3

Building a Kanban Board

1. Introduction

This lab exercise focuses on applying Agile project management principles using a Kanban board. The goal is to visualize workflow, manage tasks efficiently, and improve team productivity through work-in-progress (WIP) limits and structured task movement.

2. Objective

The objectives of this exercise are:

- To create a Kanban board for a semester project.
 - To organize tasks using defined workflow stages.
 - To implement WIP limits for better task management.
 - To track and manage User Stories developed in Week 4.
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3. Tool Used

The Kanban board was created using one of the following tools:

- Trello (*recommended for simplicity*)
- GitHub Projects (*useful for development-based projects*)

4. Kanban Board Structure

The board consists of **five columns**, each representing a stage in the workflow:

1. **Backlog**
 - Contains all project tasks and user stories not yet started.
 - Acts as a task repository.
2. **To Do**
 - Tasks selected from the backlog and ready to be worked on.
 - Prioritized based on importance.
3. **In Progress**
 - Tasks currently being worked on.
 - **WIP Limit Applied:** Maximum of **2 cards per person** to avoid overload and improve focus.
4. **QA (Quality Assurance)**
 - Tasks under testing and review.
 - Ensures that completed work meets acceptance criteria.
5. **Done**
 - Completed and fully tested tasks.
 - Represents finished work.

5. Work-In-Progress (WIP) Limit

A WIP limit was set for the **In Progress** column:

- Each team member is allowed to work on a **maximum of 2 tasks at a time**.
- This helps to:
 - Prevent multitasking inefficiencies
 - Improve task completion rate
 - Maintain quality of work

6. Integration of Week 4 User Stories

The User Stories created in Week 4 were added to the Kanban board and distributed as follows:

- Initially placed in the **Backlog**
- Moved to **To Do** based on priority
- Progressed through:
 - In Progress → QA → Done

Example User Stories:

1. *As a student, I want to search for books so that I can find resources easily.*
2. *As a librarian, I want to issue books so that users can borrow them.*
3. *As an admin, I want to manage user accounts so that access is controlled.*

Each story includes:

- Acceptance criteria
 - Task breakdown (if needed)
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7. Workflow Process

The workflow followed these steps:

1. Select tasks from the **Backlog**
 2. Move them to **To Do**
 3. Assign tasks and move to **In Progress**
 4. After completion, move to **QA** for testing
 5. Once verified, move to **Done**
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8. Benefits Observed

- Improved task visibility
 - Better team coordination
 - Reduced task overload due to WIP limits
 - Faster identification of bottlenecks
 - Enhanced workflow efficiency
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